

Machine Learning

2025/2026 Spring Trimester

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Office hours	Wednesday, 17:00 (online)
ECTS credits	6
Duration	14 weeks
Class hours	44 lecture hours and 34 practice hours
Course venue	KSE Dragon Capital Building, 3 Mykola Shpak Street, Kyiv
Course language	English

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Overview

Background and Course Rationale

Machine learning is a scientific discipline focused on creating algorithms that can identify significant patterns in large datasets without requiring explicit programming. In this course, we will explore the foundational principles and techniques of machine learning. Key topics include supervised and unsupervised learning, classification, regression, clustering, dimensionality reduction, the pitfalls of overfitting, and methods for assessing a model's generalization capabilities. To make the learning experience engaging and practical, we will implement many of these algorithms in Python, leveraging the capabilities of Google Colaboratory.

The course aims to

By the conclusion of this course, students are expected to develop a solid intuition about essential Machine Learning methods, complemented by hands-on experience. They should be capable of implementing key algorithms in machine learning, such as decision trees, backpropagation, and bagging trees when needed. Additionally, students should gain proficiency in using the scikit-learn library and clearly understand how to choose appropriate algorithms for different scenarios.

Prerequisites

Students taking this course are expected to have prior experience in programming with Python and a solid foundation in using the NumPy and Pandas libraries. Recommended prerequisite courses include "Programming Concepts" and "Data Manipulation Essentials."

Learning outcomes

By the course end, the students will be able to

- independently implement key Machine Learning methods using Python, including supervised methods (classification, regression), unsupervised methods (dimensionality reduction, clustering), regularization, basic deep learning, and ensemble learning.
- utilize the scikit-learn library to design, train, and develop reliable and robust Machine Learning models.
- explain which Machine Learning algorithms are appropriate for various scenarios.

- evaluate the performance of the algorithms using performance metrics, such as recall, precision F1, and AUC.
- read and summarize scientific papers.

Recommended literature:

1. Machine Learning by Dmytro Fishman – <https://www.slideshare.net/DimaFishman>
2. Deep Learning, Ian Goodfellow and Yoshua Bengio and Aaron Courville – <https://www.deeplearningbook.org/>
3. Pattern Recognition and Machine Learning, Christopher M. Bishop – <https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf>
4. Machine Learning by Andrew Ng – <https://www.coursera.org/learn/machine-learning>
5. Introduction to Machine Learning by Pascal Vincent given at Deep Learning Summer School, Montreal 2015 – https://videlectures.net/deeplearning2015_vincent_machine_learning/
6. Stanford CS class: Convolutional Neural Networks for Visual Recognition by Andrej Karpathy – <https://cs231n.github.io/>
7. Machine Learning Essential Concepts by Ilya Kuzovkin – <https://www.slideshare.net/iljakuzovkin>

Course Faculty



Hello there, and welcome to the Machine Learning course! I'm Roman Mishchenko, and I'm excited to be your Machine Learning course teacher.

I hold a master's degree in Data Science from the Ukrainian Catholic University. I work as a Senior Software Engineer at MacPaw, contributing to the CleanMyMac team. Additionally, I create technical news about iOS/macOS development for MacPaw Tech and DOU and have been recognized by Apple Security Research for discovering two vulnerabilities in macOS.

I am currently pursuing a PhD in Computer Science at KSE, with a focus on advanced computational methods and machine learning. My research specializes in computer vision for medical applications.

I'm excited to share this passion with you and hopefully inspire some curiosity in these areas along the way. My main goal for this course is to create a safe learning environment where students feel comfortable exploring and learning through experimentation. I encourage you to take risks, ask questions, and try new approaches without fear of making mistakes. My goal is to create a space where you can engage deeply with the material, learn from your experiences, and grow both intellectually and personally. I hope to inspire creativity, critical thinking, and a genuine passion for learning through this approach.

I'm really looking forward to getting to know all of you. Please don't hesitate to reach out if you have any questions about the syllabus or course materials. See you in class!

Denys Mykhailov



AI27 student at KSE, R&D Engineer at SoftServe

Background: I have a strong foundation in computer science and mathematics, with a focus on Machine Learning and Artificial Intelligence.

Have successfully completed this course before, and I am excited to be back as a mentor and guide for new students. Now working as an **R&D Engineer at SoftServe**, where I focus on **Generative AI (2D)** and **Agentic AI**.

It is exciting to see how the fundamental concepts, which we will discuss in this course, form the backbone of the cutting-edge autonomous systems I build daily.

Why I'm Here: I am a huge geek when it comes to the learning process. I believe that Machine Learning is the most exciting field to be in right now, and I want to share that excitement with you. ML can be tough, but it shouldn't be boring. I joined this course to help you navigate the challenges, make learning enjoyable and effective, share the cool things I see in the industry, and learn something new from you in return.

Pavlo Khomenko



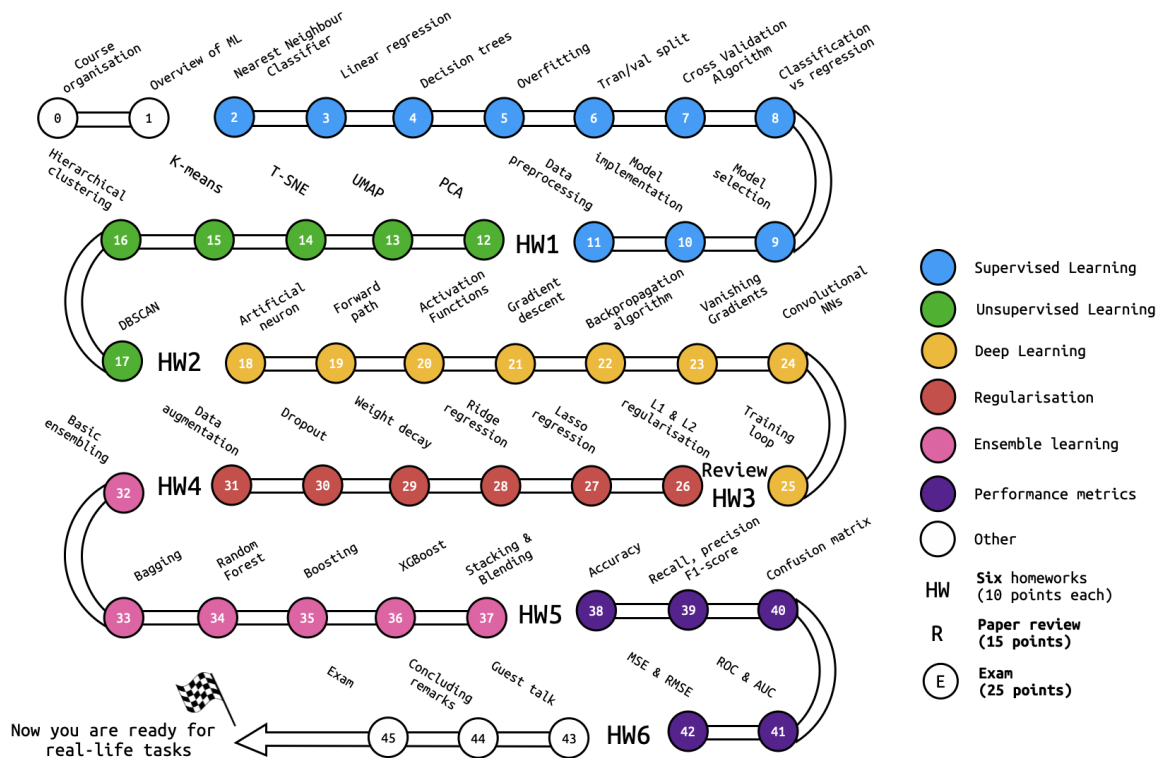
About me: I am an AI27 student at KSE and an **Applied Scientist at SoftServe**, specializing in Generative AI and NLP. My work bridges the gap between research and production engineering.

Currently, I am building **Agentic systems** to automate complex phases of the Software Development Life Cycle. Simultaneously, I am a researcher at the **KSE NLP Lab**, where I focus on fine-tuning Speech-to-Text models to solve data scarcity challenges for low-resource dialects. When I'm not training models, I enjoy jogging and bouldering, so maybe we'll meet there too.

Why I'm here: Frankly, I am just genuinely fascinated by the topics covered in this course. I deal with these concepts daily, and my goal is to share this excitement with you. I want to help you see the logic

and beauty behind the math, making complex ideas not just understandable, but truly engaging.

Course Structure



Machine Learning course includes

- 6 Modules. Each module is two weeks long (except for the first).
- 22 lectures. There will be no grading activities during lectures. Lecture materials will be published on Moodle.
- 17 practice sessions in a workshop format. These practice sessions aim to be a direct lecture continuation. The format involves completing a practical task using a suggested template simultaneously with the instructor using personal devices (laptops, etc.). During these practical sessions, students can fix any issues with the instructor and ask clarifying questions.
- 1 exam session.

Module	Topics
1 (Week 1)	Supervised Learning
	• Classification/Regression • KNN • Linear Regression • Decision Tree • Overfitting and train/validation splits • CV algorithm
2 (Week 2-3)	Unsupervised Learning
	• Data compression • The curse of dimensionality • PCA, t-SNE/UMAP (briefly) • H-clustering, K-Means, DBSCAN;
3 (Week 4-5)	Deep Learning
	• Artificial neuron • Feed-forward, backprop, convolutional operation, • Training loop
4 (Week 6-7)	Regularization methods
	• L2 and L1 regularisation • Weight decay, dropout • Data augmentation
5 (Week 8-9)	Ensemble learning
	• Basic averaging and majority vote • Bagging ensemble • Random forest • Adaptive boosting, stacking, and blending;
6 (Week 10-11)	Performance metrics
	• Accuracy, baselines • Confusion matrix, recall and precision • F1/Dice, ROC/AUC. • Holistic view

Grading Policy

Grade distribution

1. Homework assignments – 6 assignments, 10 points each, with additional tasks for extra points (32 extra points in total).
2. Paper summary – 15 points
3. Exam – 25 points

Homework assignments:

Each assignment will consist of several tasks designed to test and enhance your coding skills. For each task, students will be required to write code that produces the desired result based on the given problem statement. Clear instructions and examples will be provided to help you understand what is expected. Additionally, each assignment will include an optional task for extra points, allowing students to challenge themselves further and improve overall grades. Completing these optional tasks is encouraged, as they offer an opportunity to deepen your understanding and showcase your problem-solving abilities.

Each homework assignment must also be defended orally to the teaching assistants (TAs). During the defense, students are expected to explain their approach, justify design and implementation choices, and demonstrate understanding of the underlying concepts. **The homework grade is split evenly: 50% is awarded for the submitted notebook and 50% for the oral defense.** The defense is **mandatory**; failure to successfully defend an assignment may result in partial or no credit, even if the Moodle submission was completed on time.

Students are granted a total of **7 late days for the entire course**, which may be used flexibly across all homework assignments and are **applied to the submission deadlines in Moodle**. These days can be applied to a single assignment or split among multiple assignments; however, regardless of late-day usage, **each homework must still be defended to the teaching assistants**. The defense must take place **no later than two weeks after the original Moodle deadline**. Once all 7 late days are exhausted, no further late submissions will be accepted unless prior approval is granted due to documented exceptional circumstances.

Paper summary:

Choose one of the provided research papers, read it, and write a concise summary that captures its main contributions, key methods, results, and conclusions. The summary should highlight the main problem addressed in the paper, the approach taken to solve it, and the significance of the findings. It should also touch upon the broader context or application of the research and, if applicable, mention any challenges, limitations, or directions for future work identified by the authors. This task is designed to help develop critical reading skills and the ability to distill complex information into an accessible format.

The research paper summary should be **1-2 pages long** and will be graded based on the following criteria:

1. Content and Accuracy (30%)

- The summary accurately captures the main problem addressed, the methods used, and the key results and conclusions.
- It highlights the paper's main contributions and places them in the broader context of the field or application area.

2. Critical Insight and Reflection (25%)

- Demonstrates a clear understanding of the research significance, including challenges, limitations, and potential future directions mentioned by the authors.
- Offers a thoughtful analysis of the strengths and weaknesses of the paper, if applicable.

3. Clarity and Conciseness (25%)

- The writing is clear, focused, and concise, with no unnecessary details or jargon.
- The summary stays within the 1-2 page limit and is easy to follow, even for readers unfamiliar with the topic.

4. Structure and Presentation (20%)

- The summary is logically organized, with a smooth flow of ideas between sections or paragraphs.
- Grammar, spelling, and formatting meet professional standards, enhancing readability.

Note: Failure to stay within the 1-2 page limit, omission of key elements, or over-reliance on copying from the paper will result in point deductions. This task is an opportunity to develop critical reading and summarization skills, so originality and effort are essential.

Exam

The exam will cover all topics from the course, ensuring a thorough assessment of your understanding and ability to apply the material. It will consist of five open-ended questions, each worth 5 points, making a total of 25 points. The exam will be in a handwritten format, requiring clear and detailed explanations of your answers.

The questions will be designed to test your ability to recall key concepts, apply them to problems, and analyze or synthesize information as required. Some questions may involve solving problems or explaining processes in detail, while others may ask you to interpret scenarios or evaluate ideas.

During the exam, you are not allowed to use any external materials, such as textbooks, notes, or online resources. The use of AI tools or any form of assistance is strictly prohibited. The answers must reflect your own understanding and effort. Be sure to review all course materials thoroughly to prepare. Good luck!

Academic integrity warning:

Academic integrity is highly valued by the KSE community and its vast majority of students. We have a zero-tolerance policy towards academic plagiarism, self-plagiarism, fabrication, falsification, cheating, deception, bribery, and other types of academic integrity violations. Due to the highly competitive nature of the programs, all students must be treated equally. Even a single case of violation of academic integrity is a serious misdemeanor that may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions, and other merit-based awards. Therefore, penalties may vary from receiving zero points or a negative grade for an assignment to expulsion of the student from KSE, depending on the severity of the case, and might include additional consequences such as deprivation of scholarships, financial assistance, etc. All the rules and procedures regarding academic integrity are stated in the KSE Code of Academic Integrity. Before enrolling in this course, students must be aware of and abide by the KSE Code of Academic Integrity.

This course is designed to create a safe space for students to experiment and learn about Machine Learning. We encourage you to communicate with other students and teachers. However, all assignments (homework and paper review) are intended to showcase your individual problem-solving skills. Therefore, each task is treated as independent work unless explicitly stated otherwise. You are allowed to discuss general approaches to solving the problems with other students, but you have to do the assignments on your own.

Students who fail to adhere to these expectations will receive a failing grade on the assignment and will be notified by the teacher regarding violating the academic integrity code.

AI Tools Usage policy

It is prohibited to use AI tools (including, but not limited to, ChatGPT, Gemini, Claude, Copilot, DeepL) to write any parts of code. However, you are allowed to use AI tools for consultation or documentation search, provided that you **attach logs of your interaction with the AI tools** (e.g., chat history or screenshots). **Additionally, the use of AI tools to write texts or to fully or partially complete text-based assignments is strictly prohibited.**

Breaking those rules (such as using AI tools for code writing or not providing an artifact of work with AI tool) would carry the same consequences as cheating or plagiarizing.